COMPLEX NUMBERS

Complex numbers arose in order to solve polynomial equations such as $x^2+1=0$ or $x^2+x+1=0$, which are not satisfied by real numbers. We assume that a complex number is any number that can be written in the form $a+bi$ where a and b are real numbers and i is the imaginary unit. It has the property that $i^2=-1$. We define operations with complex numbers as follows:

1. Addition: $(a+bi)+(c+di)=(a+c)+(b+d)i$
2. Subtraction: $(a+bi)-(c+di)=(a-c)+(b-d)i$
3. Multiplication $(a+bi)(c+di)=ac+adi+bci+bdi^2$
$=(ac-bd)+(ad+bc)i$
4. Division: $\dfrac{a+bi}{c+di}=\dfrac{a+bi}{c+di}\cdot\dfrac{c-di}{c-di}$

Note that we have used the ordinary rules of algebra except that we replace i^2 by -1 whenever it occurs. The commutative associative and distributive laws apply to complex numbers

We call a, and b of $a+bi$ the real and imaginary parts respectively. Two complex numbers are equal if and only if their

real and imaginary parts are respectively equal.

SOLVED EXAMPLE

① Simplify each of the following:

ⓐ $(4-2i)+(-6+5i) = 4-6+(-2+5)i = -2+3i$

ⓑ $(-7+3i)-(2-4i) = -7-2+3i+4i = -9+7i$

ⓒ $(3-2i)(1+3i) = 3(1+3i)-2i(1+3i)$

$= 3+9i-2i-6i^2$

$= 3+9i-2i+6 = 9+7i$

ⓓ $\dfrac{-5+5i}{4-3i} = \dfrac{-5+5i}{4-3i} \cdot \dfrac{4+3i}{4+3i} = \dfrac{(-5+5i)(4+3i)}{16-9i^2}$

$= \dfrac{-20-15i+20i+15i^2}{16+9} = \dfrac{-35+5i}{25}$

$= \dfrac{5(-7+i)}{25} = \dfrac{-7+i}{5} = \dfrac{-7}{5}+\dfrac{1}{5}i$

ⓔ $\dfrac{i+i^2+i^3+i^4+i^5}{1+i} = \dfrac{i-1+(i^2)i+(i^2)^2+(i^2)^2 i}{1+i}$

$= \dfrac{i-1-i+1+i}{1+i} = \dfrac{i}{1+i}$

$= \dfrac{i}{1+i} \cdot \dfrac{1-i}{1-i} = \dfrac{i-i^2}{1-i^2} = \dfrac{i+1}{2} = \dfrac{i}{2}+\dfrac{1}{2}$

ⓕ (i) $i^{12} = (i^4)^3 = 1^3 = 1$ (ii) $i^{10} = (i^4)^2 i^2 = i^2(-1) = 1$

(iii) $i^{23} = (i^4)^5 i^3 = i^3 = -i$

TRY QUESTIONS 1

Simplify the following:

(a) $(4+7i)-(2-5i)$

(b) $(4+5i)+(3-2i)$

(c) $(5-9i)-(2-6i)+(3-4i)$

(d) $(5-4i)(5+4i)$

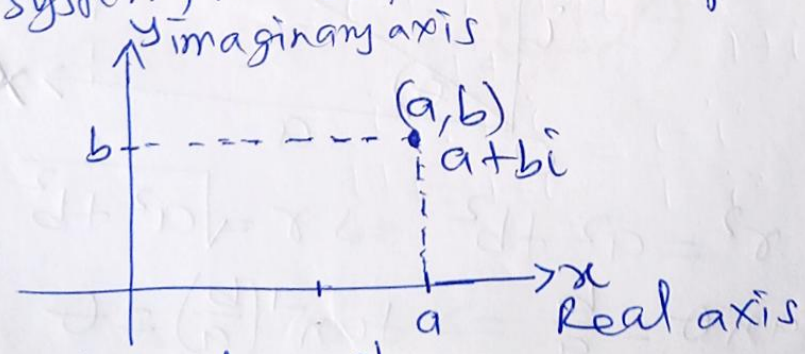
(e) $(4-3i)^2$

(f) $\frac{5-4i}{3}$

(g) $\frac{7-4i}{4+3i}$

GRAPHICAL REPRESENTATION OF COMPLEX NUMBERS

The complex number $a+bi$ is a unique ordered pair of real numbers (a,b), and vice versa. For instance $3-5i$ corresponds to $(3,-5)$. Associating these ordered pairs ~~with~~ of real numbers with points in a rectangular coordinate system, we obtain a complex plane.



Complex plane.

When complex numbers are associated with points in a rectangular coordinate system, we refer to the x-axis as the real axis and the y-axis as the imaginary axis. The complex number $a+bi$ is said to be in rectangular form.

Try Question 2

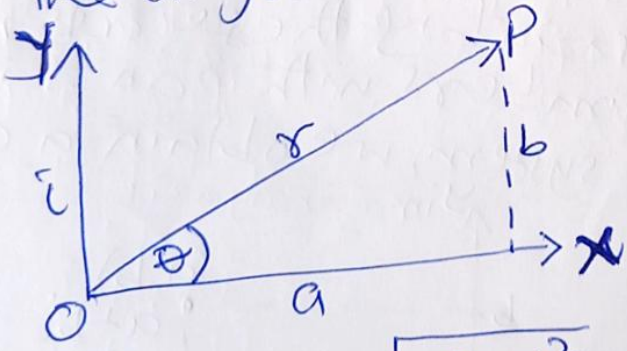
Plot the following complex numbers in a complex plane:

(a) $A = 2+3i$, $B = -3+5i$, $C = -4$, $D = -3i$

(b) $A = 4+2i$, $B = 2-3i$, $C = -5$, $D = 4i$

POLAR FORM OF A COMPLEX NUMBER [from rectangular to Polar]

A complex number ~~$z = a+bi$~~ $z = x+iy$ can be considered as a point P, with coordinates (x, y) on a rectangular xy plane called in this case the complex plane or Argand diagram. On an Argand diagram, let OP be a vector $a+bi$. Let r = length of vector and θ the angle made with OX.



Then $r^2 = a^2 + b^2 \Rightarrow r = \sqrt{a^2+b^2}$

and $\tan\theta = \frac{b}{a} \Rightarrow \tan^{-1}\left(\frac{b}{a}\right) = \theta$

Also $a = r\cos\theta$ and $b = r\sin\theta$

Since $z = a + bi$, this can be written as

$z = r\cos\theta + ir\sin\theta$ ie $z = r(\cos\theta + i\sin\theta)$

This is called the polar form of the complex number $a+bi$, where

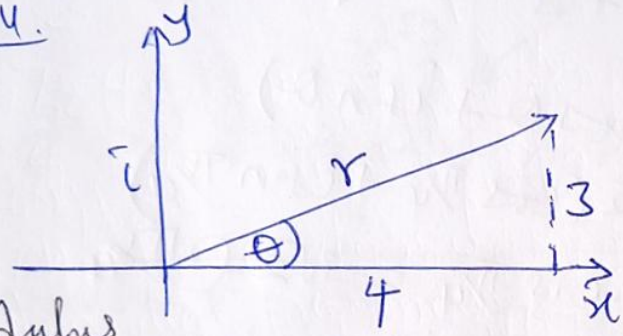
$$r = \sqrt{a^2+b^2} \text{ and } \theta = \tan^{-1}\left(\frac{b}{a}\right)$$

Also $z = x + yi = r(\cos\theta + i\sin\theta) = re^{i\theta}$

SOLVED PROBLEMS 2

1. Express $z = 4+3i$ in polar form

Solu.



Modulus

(a) $r^2 = 4^2 + 3^2 = 16 + 9 = 25$

$r = 5$

Amplitude

(b) $\tan\theta = \frac{3}{4} = 0.75$

Amplitude $\theta = 36^\circ.52'$

$\Rightarrow z = 5(\cos 36^\circ.52' + i\sin 36^\circ.52')$

Q2 Express the following in Polar form (6)

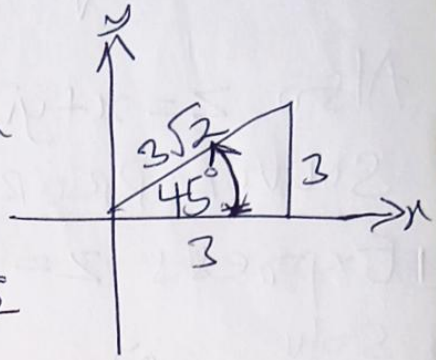
(a) $3+3i$

(b) $-1+\sqrt{3}i$

(c) -1

(d) $-2-2\sqrt{3}i$

SOLUTION



(a) Amplitude $\theta = 45° = \pi/4$ radian

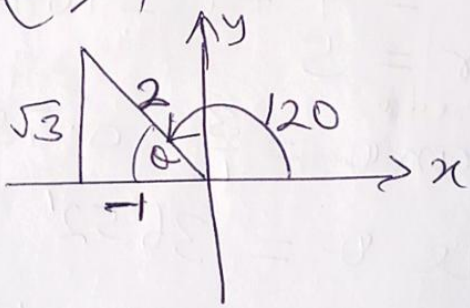
(ii) Modulus $r = \sqrt{3^2+3^2} = 3\sqrt{2}$

then $3+3i = r(\cos\theta + i\sin\theta)$

$= 3\sqrt{2}(\cos\pi/4 + i\sin\pi/4)$

$= 3\sqrt{2}\,\text{cis}\,\pi/4$ ~~$= 3\sqrt{2}\,e^{\pi i/4}$~~

$= 3\sqrt{2}\,e^{\pi i/4}$



(b) $-1+\sqrt{3}i$

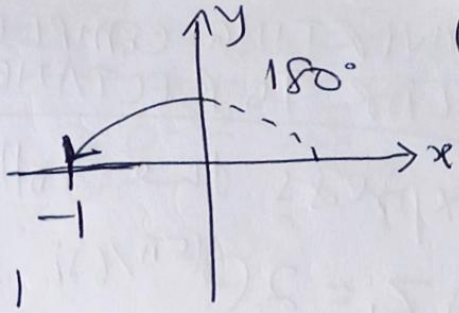
(i) Amplitude $\theta = 120°$

$= \frac{2\pi}{3}$ rad

(ii) modulus $r = \sqrt{(-1)^2+(\sqrt{3})^2} = \sqrt{1+3} = 2$

Then $-1+\sqrt{3}i = 2(\cos(2\pi/3) + i\sin(2\pi/3))$

(c) -1



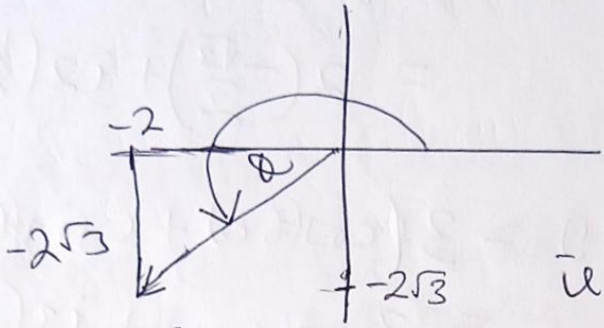
(i) Amplitude $\theta = 180° = \pi$ rad.

(ii) Modulus $r = \sqrt{(-1)^2 + 0^2} = 1$

then $-1 = 1(\cos\pi + i\sin\pi)$

$= \text{cis}\,\pi = e^{\pi i}$

(d) $-2-2\sqrt{3}i$



ie $\tan\theta = \frac{-2\sqrt{3}}{-2}$

$= \sqrt{3}$

$\theta = \tan^{-1}\sqrt{3})$

$= 60°$

Arg $= 60 + 180°$

$= 240°$

(i) Amplitude $\theta = 240° = 4\pi/3$ rad

(ii) Modulus $r = \sqrt{(-2)^2 + (-2\sqrt{3})^2} = 4$

$-2-2\sqrt{3} = 4(\cos 4\pi/3 + i\sin 4\pi/3)$

$= 4\,\text{cis}\,4\pi/3 = 4e^{4\pi i/3}$

TRY QUESTIONS

Express the following in polar form.

(a) $-\sqrt{3} + i = z_1$

(b) $1 - i = z_2$

(c) $-5 - 2i = z_3$

(d) $1 + i\sqrt{3} = z_4$

CONVERTING COMPLEX NUMBERS FROM POLAR TO RECTANGULAR FORM

Express the following to rectangular form

(a) $z_1 = 2e^{(5\pi/6)i}$ (b) $z_2 = 3e^{(-60^\circ)i}$ (c) $7.19e^{-2.13i}$

Note: $5\pi/6 = 150^\circ$

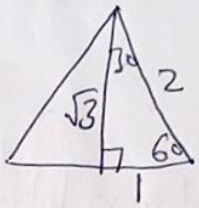
Solution

(a) Let $x + iy = 2e^{(5\pi/6)i}$

$$= 2\left(\cos(5\pi/6) + i\sin(5\pi/6)\right)$$

$$= 2\left(-\frac{\sqrt{3}}{2}\right) + i2\left(\frac{1}{2}\right) = -\sqrt{3} + i$$

(b) $x + iy = 3(\cos(60) + i\sin(60))$



$$= 3\cos(60) + i3(\sin 60)$$

$$= 3\left(\frac{1}{2}\right) + i3\left(\frac{\sqrt{3}}{2}\right)$$

$$= \frac{3}{2} + \frac{3\sqrt{3}}{2}i$$

(c) $x + iy = 7.19e^{-2.13i}$

$$= 7.19(\cos(-2.13) + i\sin(-2.13))$$

$$= -3.81 - 6.09i$$

TRY QUESTIONS

Express the following in rectangular form

(a) $z_1 = \sqrt{2}e^{(-\pi/2)i}$

(b) $z_2 = 3e^{120^\circ i}$

(c) $z_3 = 6.49e^{-2.08i}$

Products and Quotients in Polar Form

Theorem 1

If $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$, then

1. $z_1 z_2 = r_1 e^{i\theta_1} \cdot r_2 e^{i\theta_2} = r_1 r_2 e^{i(\theta_1+\theta_2)}$

2. $\dfrac{z_1}{z_2} = \dfrac{r_1 e^{i\theta_1}}{r_2 e^{i\theta_2}} = \dfrac{r_1}{r_2} e^{i(\theta_1-\theta_2)}$

Proves:

1. $z_1 z_2 = r_1 e^{i\theta_1} r_2 e^{i\theta_2}$

$$= r_1 r_2 (\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2)$$

$$= r_1 r_2 (\cos\theta_1\cos\theta_2 + i\cos\theta_1\sin\theta_2 + i\sin\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2)$$

$$= r_1 r_2 [(\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2) + i(\cos\theta_1\sin\theta_2 + \sin\theta_1\cos\theta_2)]$$ Sum identity

$$= r_1 r_2 [\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)]$$

$$= r_1 r_2 e^{i(\theta_1+\theta_2)}$$

2. $\dfrac{z_1}{z_2} = \dfrac{r_1(\cos\theta_1 + i\sin\theta_1)}{r_2(\cos\theta_2 + i\sin\theta_2)} = \dfrac{r_1(\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 - i\sin\theta_2)}{r_2(\cos\theta_2 + i\sin\theta_2)(\cos\theta_2 - i\sin\theta_2)}$

$$= \frac{r_1}{r_2} \frac{(\cos\theta_1\cos\theta_2 + i\sin\theta_1\cos\theta_2 - i\cos\theta_1\sin\theta_2 + \sin\theta_1\sin\theta_2)}{(\cos^2\theta_2 + \sin^2\theta_2)}$$

$$= \frac{r_1}{r_2}[(\cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2) + i(\sin\theta_1\cos\theta_2 - \cos\theta_1\sin\theta_2)]$$

$$= \frac{r_1}{r_2}[\cos(\theta_1-\theta_2) + i\sin(\theta_1-\theta_2)] = \frac{r_1}{r_2} e^{i(\theta_1-\theta_2)}$$

Que 1

If $z_1 = 8e^{45°i}$ and $z_2 = 2e^{30°i}$ then

(a) $z_1 z_2 = 8e^{45°i} \cdot 2e^{30°i} = 2 \cdot 8\, e^{i(45°+30°)} = 16e^{75°i}$

(b) $\frac{z_1}{z_2} = \frac{8e^{45°i}}{2e^{30°i}} = \frac{8}{2} e^{i(45°-30°)} = 4e^{15°i}$

(2) Evaluate the ff

(a) $\frac{6(\cos 72° + i\sin 72°)}{2(\cos 41° + i\sin 41°)}$

(b) If $z_1 = 8(\cos 65° + i\sin 65°)$ and $z_2 = 4(\cos 27° + i\sin 27°)$. Find (i) $z_1 z_2$ (ii) $\frac{z_1}{z_2}$

Solution

(a) $\frac{6}{2}\frac{(\cos 72° + i\sin 72°)}{(\cos 41° + i\sin 41°)} = 3(\cos 31° + i\sin 31°)$

(b) (i) $z_1 z_2 = 32(\cos 88° + i\sin 88°)$

(ii) $\frac{z_1}{z_2} = 2(\cos 42° + i\sin 42°)$

TRY QUESTIONS

(1) If $z_1 = 9e^{165°i}$ and $z_2 = 3e^{55°i}$, find

(a) $z_1 z_2$ and (b) z_1/z_2

Q3 Simplify the following

(a) $3(\cos 143^\circ + i\sin 143^\circ) \times 4(\cos 57^\circ + i\sin 57^\circ)$

(b) $\dfrac{10(\cos 126^\circ + i\sin 126^\circ)}{2(\cos 72^\circ + i\sin 72^\circ)}$

If z_1 and z_2 are complex numbers, prove that (1) $|z_1 z_2| = |z_1||z_2|$

Solution

Let $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$. Then

$|z_1 z_2| = |(x_1+iy_1)(x_2+iy_2)| = |(x_2x_1 - y_1y_2 + i(x_1y_2 + x_2y_1)|$

$= \sqrt{(x_1x_2 - y_1y_2)^2 + (x_1y_2 + x_2y_1)^2}$

$= \sqrt{x_1^2x_2^2 + y_1^2y_2^2 + x_1^2y_2^2 + x_2^2y_1^2}$

$= \sqrt{(x_1^2+y_1^2)(x_2^2+y_2^2)} = \sqrt{x_1^2+y_1^2}\sqrt{x_2^2+y_2^2}$

$= |x_1 + iy_1||x_2 + iy_2| = |z_1||z_2|$

(2) $|3-4i||3+4i| = \sqrt{(3)^2+(-4)^2}\sqrt{(4)^2+(3)^2} = 25$

(3) $\left|\dfrac{1}{1+3i} - \dfrac{1}{1-3i}\right| = \left|\dfrac{1-3i}{1-9i^2} - \dfrac{1-3i}{1-9i^2}\right| = \left|\dfrac{-6i}{10}\right|$

(4) Solve $x^3 - 2x - 4 = 0$

Soln The possible roots are $\pm 1, \pm 2, \pm 4$

$(x-2)(x^2+2x+2) = 0$

Using $x = \dfrac{-b \pm \sqrt{b^2-4ac}}{2a}$, we have

$\dfrac{-2 \pm \sqrt{4-8}}{2} = \dfrac{-2+2i}{2} = -1 \pm i$

DE MOIVRE'S THEOREM

If $z = x + iy = re^{i\theta}$, and n is a natural number, then

$$z^n = (x+iy)^n = (re^{i\theta})^n = r^n e^{n\theta i}$$

Qne 1

Use De Moivre's theorem to find $(1+i)^{10}$. Write the answer in exact rectangular form.

Solution

$i+1$

$r = \sqrt{1^2+1^2} = \sqrt{2}$

Arg: $\tan\theta = 1$

$\Rightarrow \theta = \tan^{-1}(1) = 45^\circ$

$i+1 = re^{i\theta} = \sqrt{2}e^{45^\circ i}$ Convert $i+1$ to polar form

$z^n = r^n(e^{i\theta})^n = r^n e^{i\theta n}$ Use De Moivre's theorem

$(i+1)^{10} = (\sqrt{2})^{10}(e^{45\times 10})^i = 2^5 e^{450i}$

$= 32e^{450i}$

$= 32(\cos 450^\circ + i\sin 450^\circ)$ change to rectangular form

$= 32(0+i) = 32i$

Q2. Use De Moivres theorem to find $(-\sqrt{3}+i)^6$. Write the answer in exact rectangular form.

Solution

$$(-\sqrt{3}+i)^6 = (2e^{150j})^6 i$$
$$= 2^6 e^{150(6)i}$$
$$= 64e^{900i}$$
$$= 64(\cos 900 + i\sin 900^\circ)$$
$$= 64(-1+0i) = \underline{\underline{-64}}$$

TRY QUESTIONS

Use De Moivre's theorem to find the following (a) and (b). Write your answers in exact polar and rectangular forms.

(a) $(1+i\sqrt{3})^5$

(b) $(-i\sqrt{3}+1)^4$